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# Problem 6: Perm Check

```
FUNCTION PermCheck(A):
    N ← length of A
    Seen ← array of (N+1) booleans,
whereby are all false

    FOR each value IN A:
        IF value > N THEN
            RETURN 0
        IF seen[value] = TRUE THEN
            RETURN 0
        seen[value] ← TRUE
    RETURN 1
```

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# Problem 7: FrogRiverOne

```
FUNCTION FrogRiverOne(X, A):
    Covered ← array of (X+1) booleans, all FALSE
    remaining ← X

    FOR time ← 0 TO length(A)-1:
        position ← A[time]
        IF position ← X AND Covered[position] = FALSE THEN
            Covered[position] ← TRUE
            remaining ← remaining - 1
            IF remaining = 0 THEN
                RETURN time
    RETURN -1
```

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# Problem 8. Passing Cars

```
FUNCTION PassingCars(A):
    LIMIT ← 1,000,000,000
    east ← 0
    pairs ← 0

    FOR each direction IN A:
        IF direction = 0 THEN
            east ← east + 1
        ELSE
            pairs ← pairs + east

            IF pairs > LIMIT THEN
                RETURN -1

    RETURN pairs
```

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# Problem 9: MissingInteger

```
FUNCTION MissingInteger(A):
    N ← length of A
    Seen ← array of (N+2) booleans, all FALSE
    FOR each value IN A:
        IF value >= 1 AND value <= N THEN
            Seen[value] ← TRUE
    FOR candidate ← 1 TO N+1:
        IF Seen[candidate] = FALSE THEN
            RETURN candidate
```

# Problem 10: CountDiv

```
FUNCTION CountDiv(A, B, K):
    Count ← floor(B/K) - floor(A/K)
    IF A MOD K = 0 THEN
        Count ← Count + 1
    RETURN Count
```